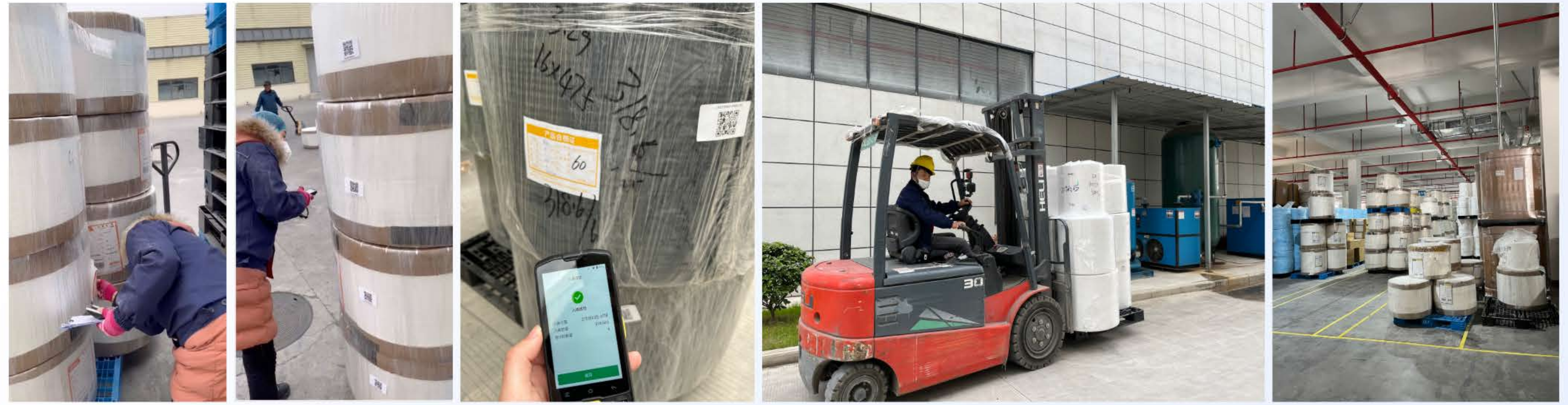
User Experience Optimization of WMS Execution



A warehouse worker (XiaJie) input 2000 rolls raw materials (non-woven fabric) by Blacklake 3.0 WMS with optimized user experience saved 2 hours.

Frontline workers

Scheduler

PDA

WEB

ShangHai · 2023

Black Lake Technologies × U-PLAY

01 Background

Roadmap of 930 Campaign

Priority of iteration

P0: User experience optimization (focusing on addressing system operational efficiency issues)

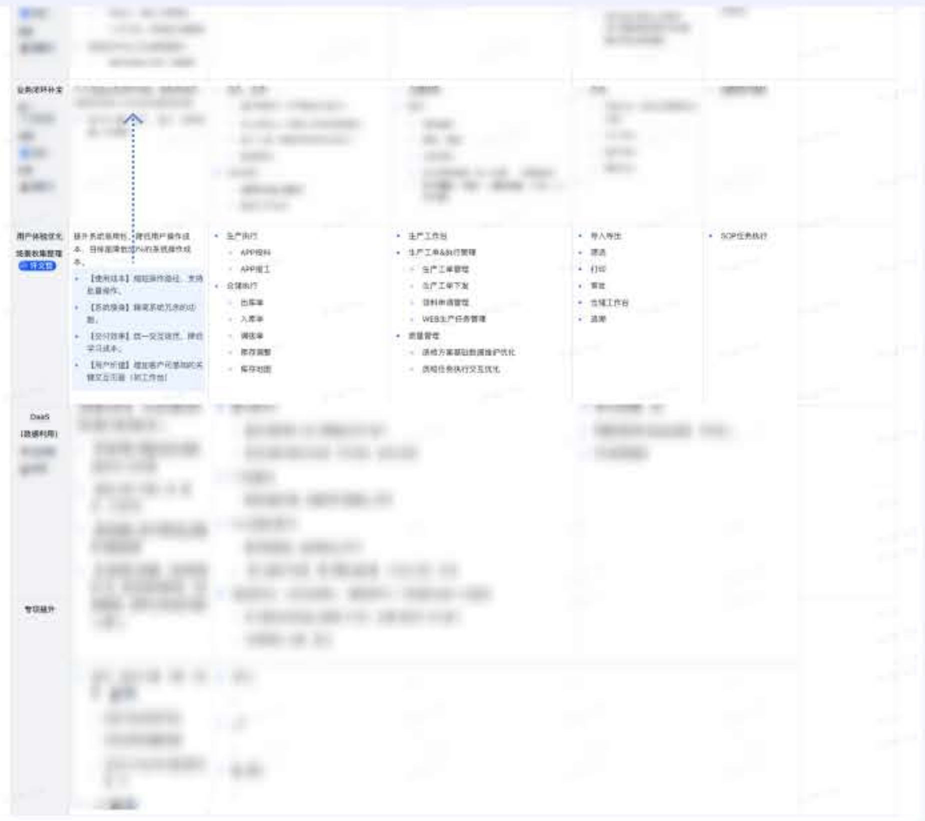
P1: Customized development support (with the ability to solve [illegible]0% of customized needs)

P2: Industry Characteristics and Business Loop Function Iteration

P3: Daas and Open Platforms

Targets of User Experience Design Group:

【Time Saving】Shorten the operation path, support batch operations.
【System Slimming 】Simplify the redundant functions of the system.
【Learning Efficiency】Unified interaction standards to reduce learning costs.
【User Value】Increase key interactive pages (data panel) that customers can perceive.



At the beginning of the new fiscal year 2023, intelligent manufacturing (IM) business line has formulated a 930 campaign, the general manager of IM business line set up 5 Special Groups.
The User Experience Design Special Group, as a P0 level, has been established separately.
Intended to improve system usability, reduce user operating costs, and lower system operating costs by 50%.

My role

Group Manager

UX & UI Designer

Outcome：Input time for 1 roll

A return visit to Xia Jie, a warehouse worker at U-Play, she XiaJie need to input 2000 rolls raw materials (non-woven fabric) by Balcklake 3.0 system everyday,
saved her 2 hours of operation time every day

2'24'

1'04"

Before

1. Find the task - scan the code into 5 clicks (click 4 times to enter the scanning page, click 1 time to submit after scanning)；

2. Complete one input action: 2N clicks (N is the number of codes) (call keyboard (point 1) - enter - collapse keyboard point (point 1)) Keep sliding to find the data input box

After

1. Find the task - scan the code into 2 clicks；

2. Complete one input action: N clicks (N is the number of codes) (premise: After the PDA scans the code in, it directly calls the keyboard

02 Making Planning

"Double Diamond Design Model"

-a design process model created by the British Design Council that visualizes how designers can tackle complex challenges by breaking them into four phases: Discover, Define, Develop, and Deliver

Taking Q1 (April-June) of FY 2023 as an example for warehousing _ 17th.Mar.2023

	Designing the right thing		Designing things right			
	Empathize	Define	Ideate	Build, test, iterate	Test	Delivery
*outcome file	Discover pain point issues collection forms, mind maps, questionnaire results, current situation (operation time, steps, etc.) Measurement standards can be pre-set (@ PM generates relevant dimensions based on business scenarios)	Synthesis pain point Mind maps、forms	Ideation PRD、Design	Prototype Development implementation Design、test walkthrough	Internal evaluation Docs. or forms	Delivery and promotion 1、Write Guidebook (@Wenzhe) 2、Collect feedback from users (Return to the site)

Empathize — Collect — Insight — Collect — Feedback

Business function process optimization

Empathize:
- PM (@Peng)
- **UED (@Wenzhe Xu)**
- Test engineer (@Yuan)
- CS

Collect: UED

Insight:
- PM (@Peng)
- UED (@Wenzhe Xu)
- Test engineer (@Yuan)
- CS

Collect: **UED (@Wenzhe Xu)**

Milestone meeting @Group all

Milestone meeting @Group all

- PM (@Peng)
- **UED (@Wenzhe Xu)**

PM/UED/ Front-end/ Back-end/ Test engineer

Milestone meeting @Group all

Measurement standards
Based on the dimensions of the March

UED @Group all

Milestone meeting @Group all

@Group all

Test and score according to standards

Standardization consistency optimization

- UED (@Mika)
- Front-end (@Lei)
- PM (@Shawn)
- Test Engineer (@Xiang)

UED

- UED (@Mika)
- Front-end (@Lei)
- PM (@Shawn)
- Test Engineer (@YiNa)

UED (@Mika)

Method

First-hand research
field research

Desk research
interview CS (Customer success) 、CS need to collect usability issues of the system in a form
UED use system frequently and record usability issues.

to do:
Preparation work before going to the site
1. Which venues should we go to? @Li Chun(Mananger of CS(customer success)) (deep users: eg: U-Play, at least 3-4 manufactories)
2. The conversion issue of Weipeng(customer success) and Xiaojun(customer success) . @UED-ALL @ Liu Xiao(PM) @ Pengfei(PM)
3. Prepare your own questions (JIRA, self use) @ Project Team ALL

then, Organize everyone to align

03 Planning of deliverables

UED Group- Planing

	Phase 1: Identify design opportunity points	Phase 2: Output & Implementation	Phase 3: Internal evaluation adjustment & external promotion
To Do	1. Develop quantitative standards for user experience 2. Collect issues (Using 3.0 System、Factory research, telephone follow-up, etc.) ◦ step1 ▪ Use the 3.0 system on your own ▪ Browse JIRA ▪ Conversion issues of Weipeng and Xiaojun ▪ Create a survey questionnaire ▪ Collection of CS experience issues ◦ step2 ▪ Factory research ▪ distribute questionnaires By WJX	1. Generate requirements by JIRA 2. Implement solutions based on JIRA's requirement pool	1. Internal ◦ Use-This optimized feature ◦ Rating-(based on quantitative standards of user experience) 2. External ◦ Delivery ◦ Highlight the optimization points of this presentation (text - publish on various BBS, conference - experience optimization presentation) ◦ Collect feedback (scoring sheet, on-site visit)
Key deliverables	1. Issues collection forms ◦ Rolling collection and discussion of questions from team members before on-site visit ◦ Research records of U-play by Wen Zhe, Lei and Moki ◦ CS collection checklist (registration) ◦ Issues collection meeting 2. Experience Optimization Point Range 3. The current usage data that needs to be optimized points ◦ Task operation efficiency ◦ Task completion rate ◦ Task completion time ◦ Function utilization rate 4. Questionnaires	1. JIRA 2. Product PRD (including burial plan) 3. Design draft 4. Feature Development 5. Test	1. Optimization point document for this quarter 2. Promotion Conference 3. Collect authentic user feedback documents

Related deliverables(Raw planning docs.)

04 Making Standard

Research mainstream measurement standards

Google PULSE

Ali-Cloud USE

Anti-Design PTECH

Abstract standards for BlackLake MES

1. Easy to Use (Task completion rate, Task completion time)
2. Easy to Read (Clear)

- Usability
  - Usability measurement scoring
- Satisfaction
  - NPS rating
- Consistency
  - Product consistency differences
    - Information architecture, framework, page structure
- Task Efficiency
  - Task operation efficiency
  - Task completion rate
  - Task completion time
  - Function utilization rate
  - etc
- Performance
  - Page load time
  - Network request time
  - etc

05 Confirming optimization points

Collect issues

Persona

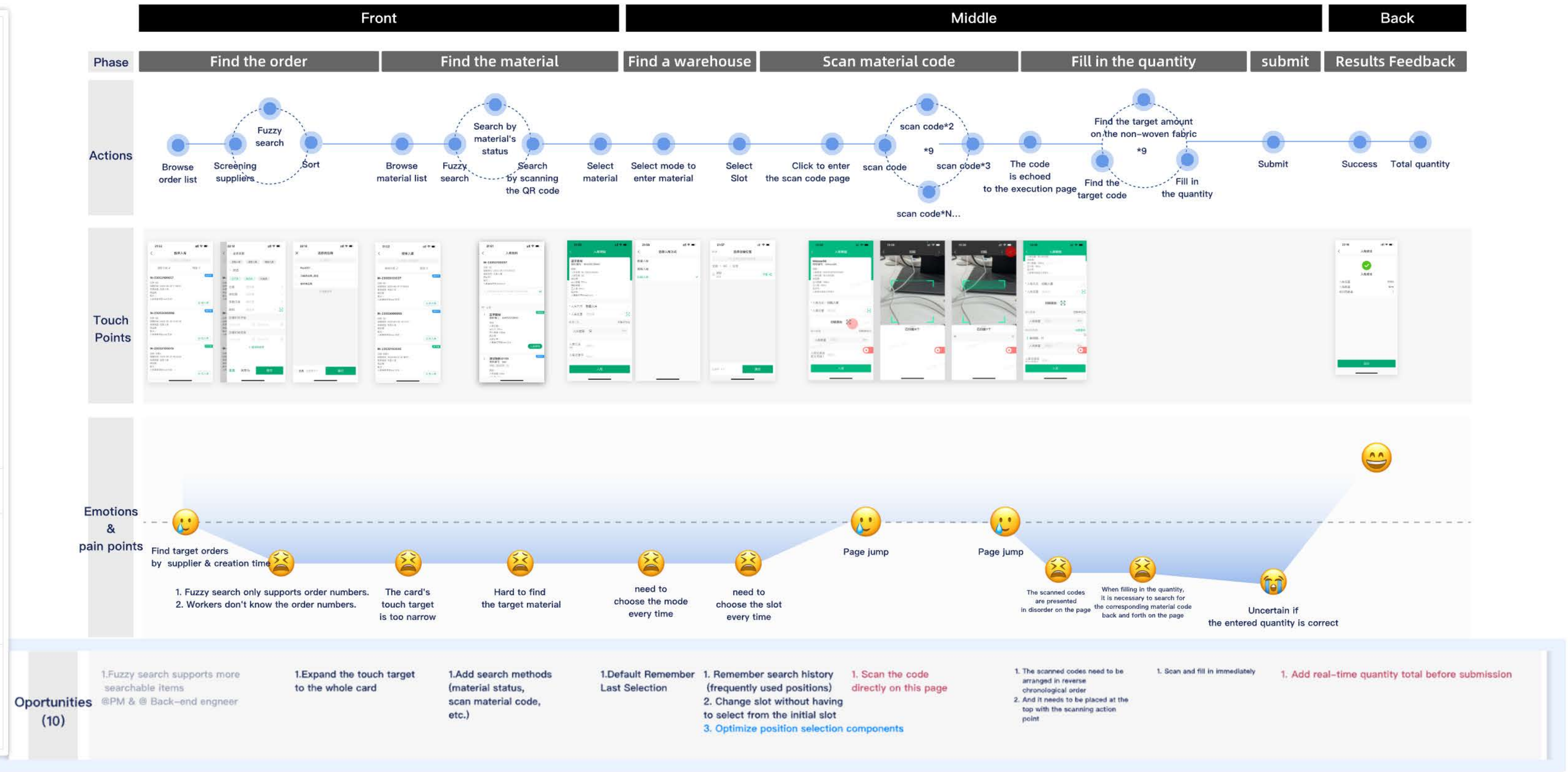
Name: XiaJie | Age:35 | Gender: F | Position: Warehouse clerk

Job Description:

Enter the data of 2000 rolls of non-woven fabric (200 pallets, 9-12 rolls per pallet) into the system by scanning code mode.(3 modes: code, non-code, Post-generation code)

Job characteristics:

Large quantity, high accuracy required, using PDA for scanning and entering codes



06 Scanning Mode

The on-site scene is that the hardware devices used by workers is: PDA

However, because the advantages of PDA mode were not considered during system development, workers were uniformly treated as using mobile phone mode (no scanner interface) and were directed to a new QR code scanning page.


Step1


Step2


Step3


Step4

Step1:
200 pallets everyday, 9-12 rolls per pallet

Step2:
Enter the scanning page 👉 scan the code in sequence

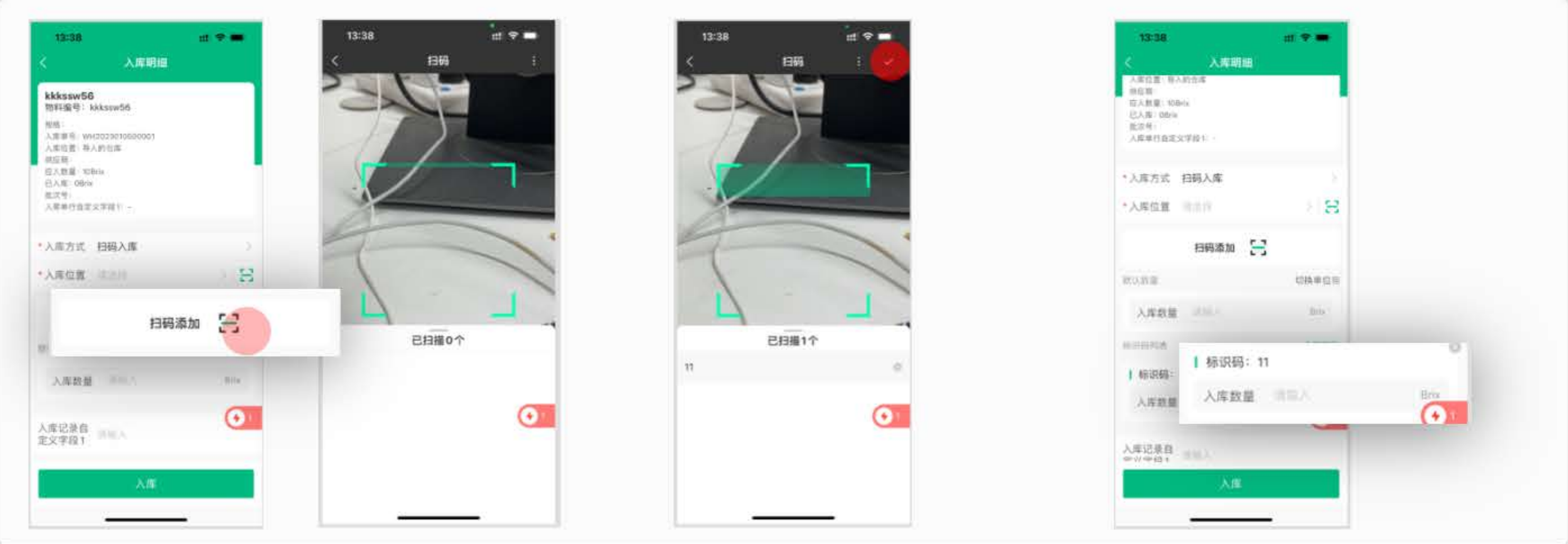
Step3:
Check to prevent missing scans
Need to repeatedly compare and fill in, with a high error rate.

Step4:
Fill in the quantity based on the quantity label on each roll

⚠️ Causing workers to need to enter a new page when scanning the code,
At the same time, the function supports continuous scanning of codes,
resulting in the display of 10 blank data to be filled in.

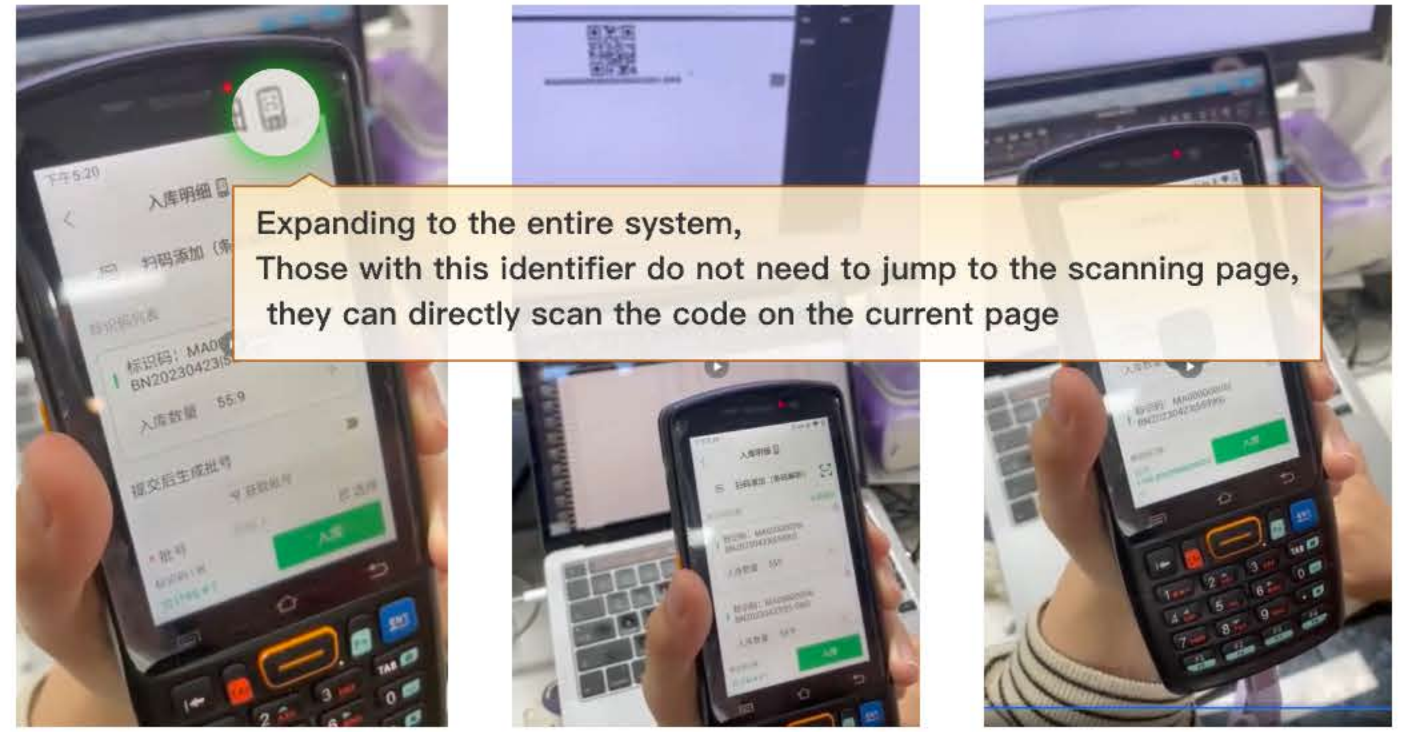
Before User: 2N clicks, 2N jumps (N=number of identification codes)

Click the scan btn → Enter the scan page → Scan the Code → Submit → Jump to the operation page



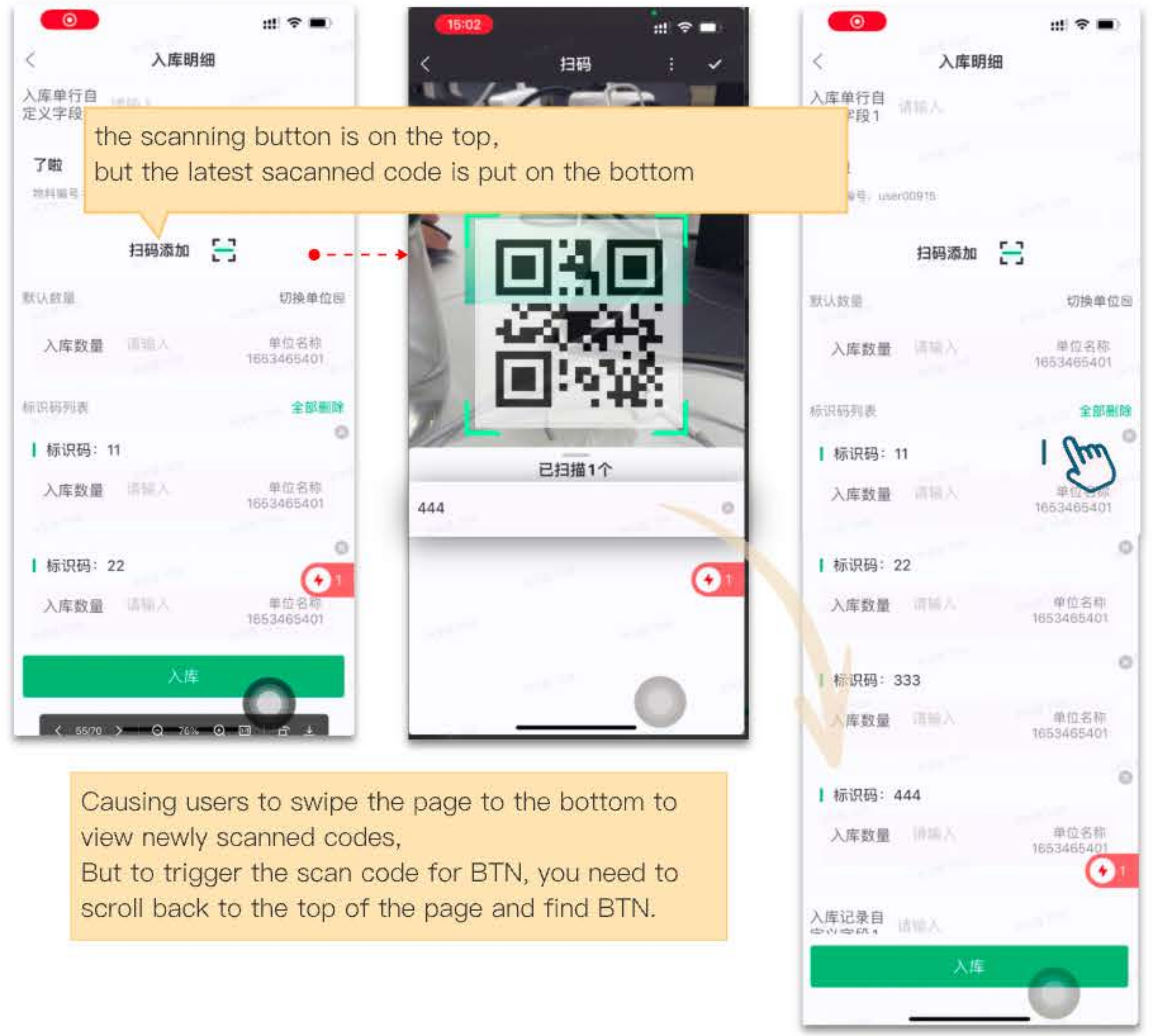
After User: 0 clicks, 0 jumps（on this Current page）

Scan the first code → Scan the 2nd code → Scan the 3rd code



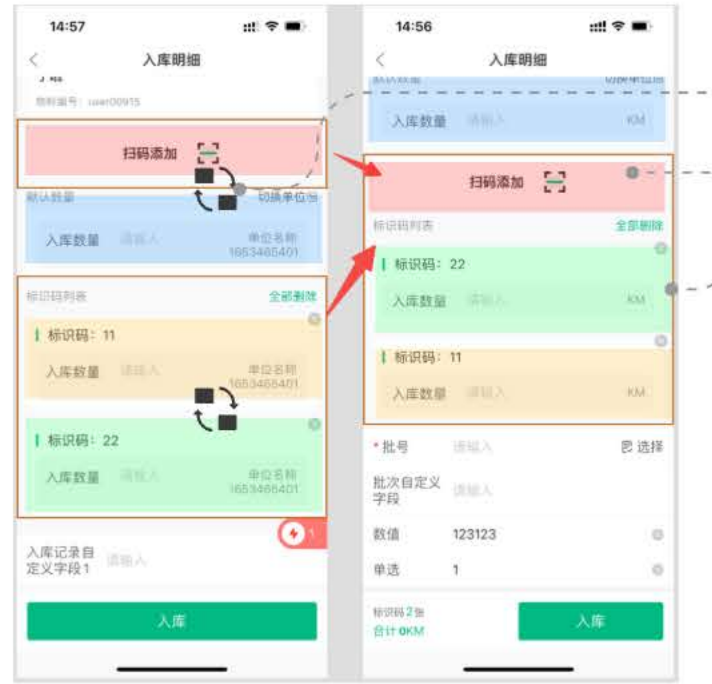
07 Information Layout

Before



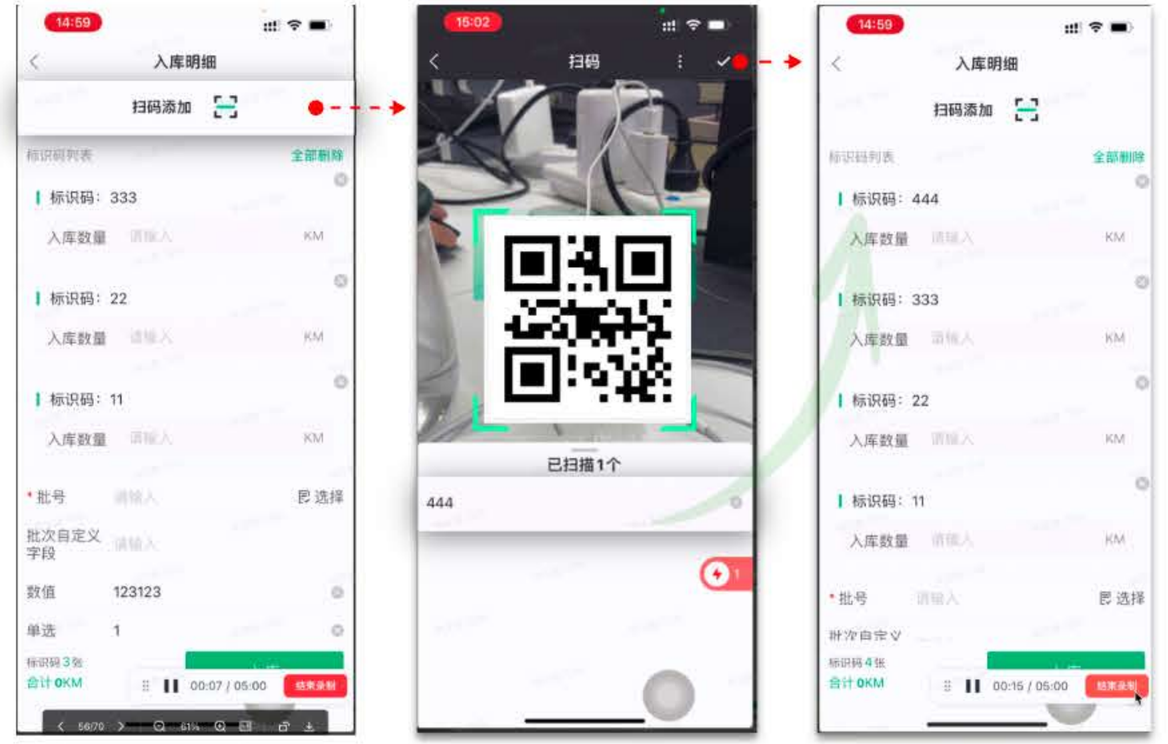
After：Put the scanning button and the latest scanned code together

Layout



- Swap the positions of scanning btn and default amount
- Arrange the scanning btn closely to the code list
- Put the latest sacanned code is put on the top

Interactive Links



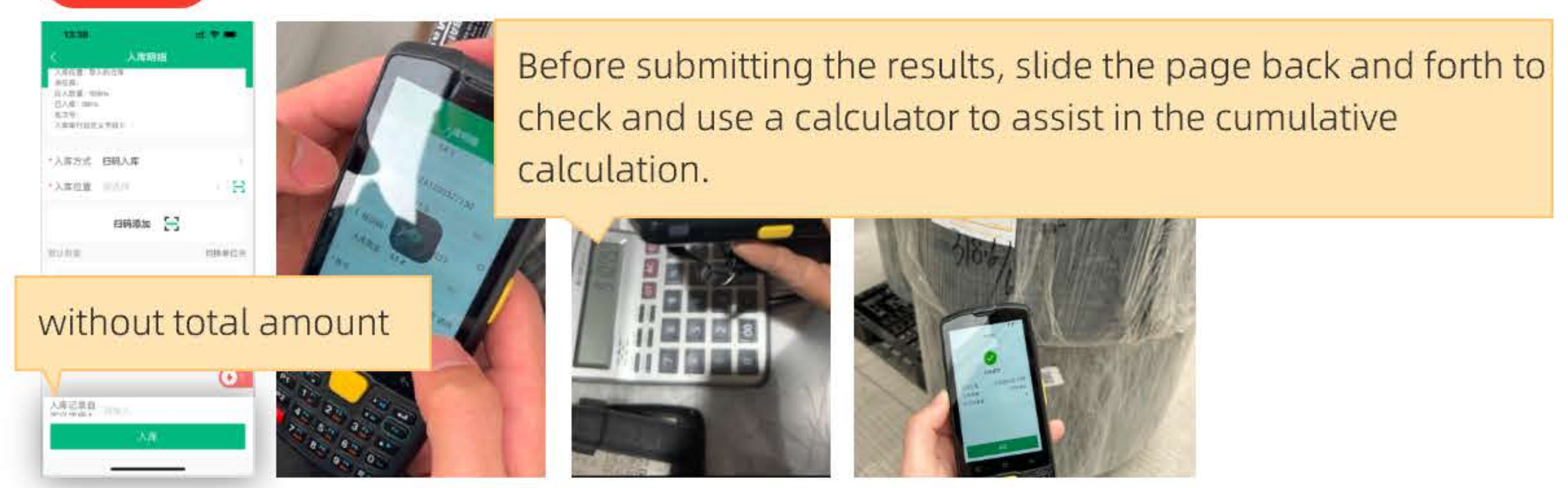
On-machine effect

08 Total amount

The business scenario is:

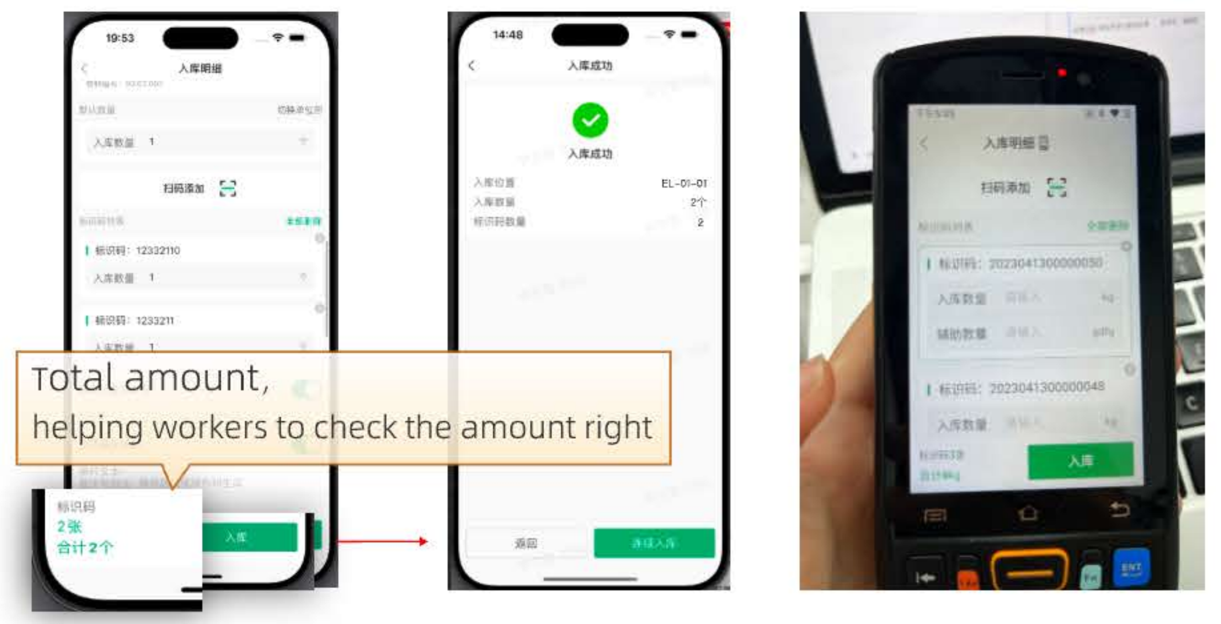
Workers cannot verify whether the total weight of non-woven fabric is correct before submission,
Workers know the total quantitynly after submission, coursing error correction costs and affecting efficiency.

Before



After

On-machine effect



09 Multimodal feedback

When the hardware devices used by workers is: Mobil Phone

Pain Point

When scanning with a mobile phone, there's no built-in feedback from the PDA hardware: beeps and vibrations.
This means workers can't tell whether the code has been successfully recognized.

Solution

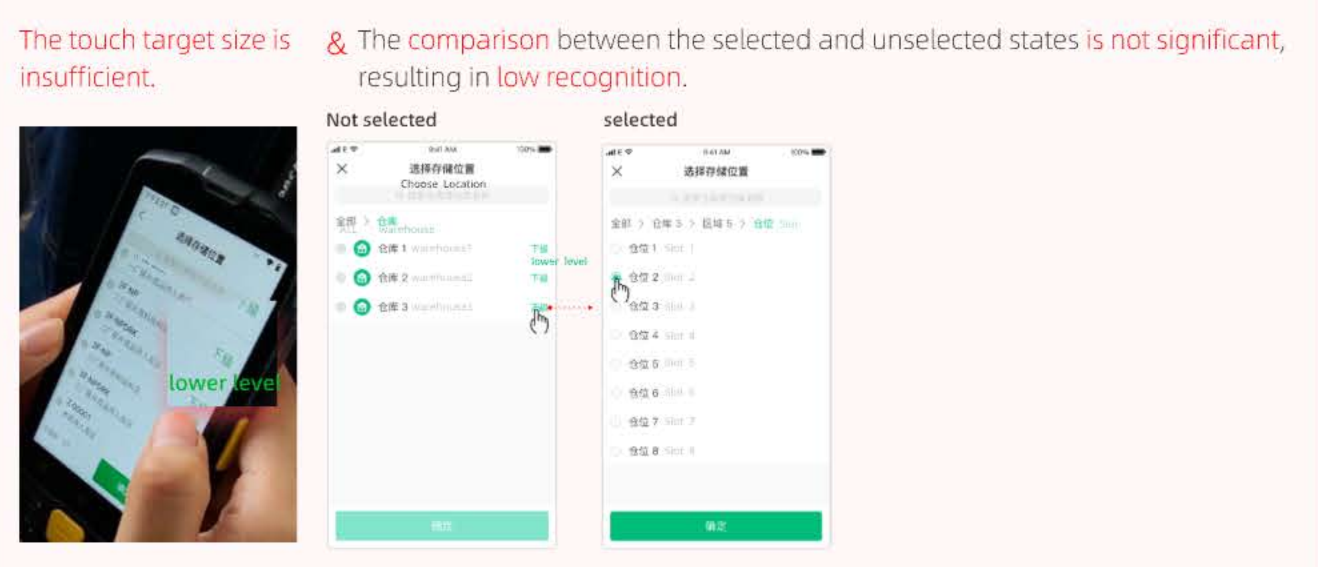
Add corresponding **physical feedback: Sound, Vibration**

- Search open-source audio sources👉 download materials👉 test clarity

- Add global configuration on the app to configure the scanning feedback for this hardware device.

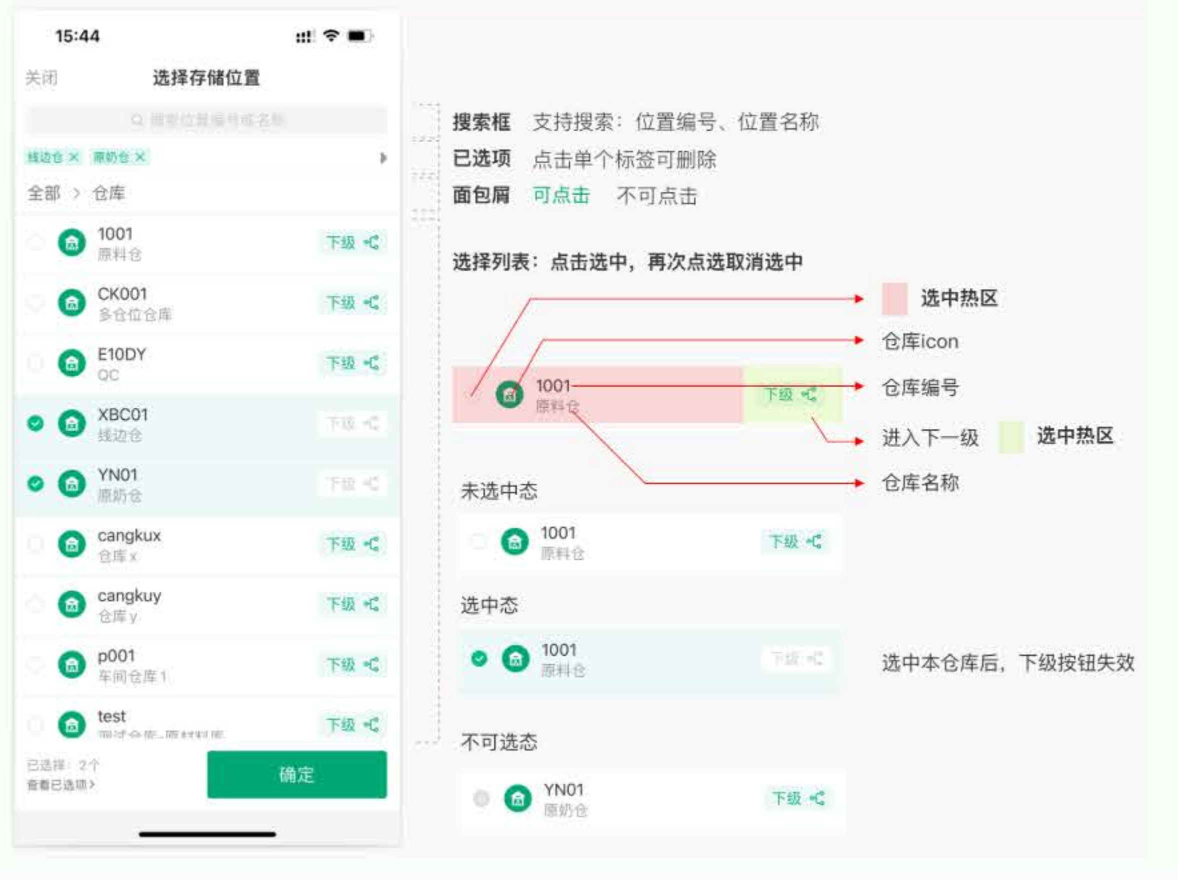
10 Optimize components

Before (Clarity and recognition issues)

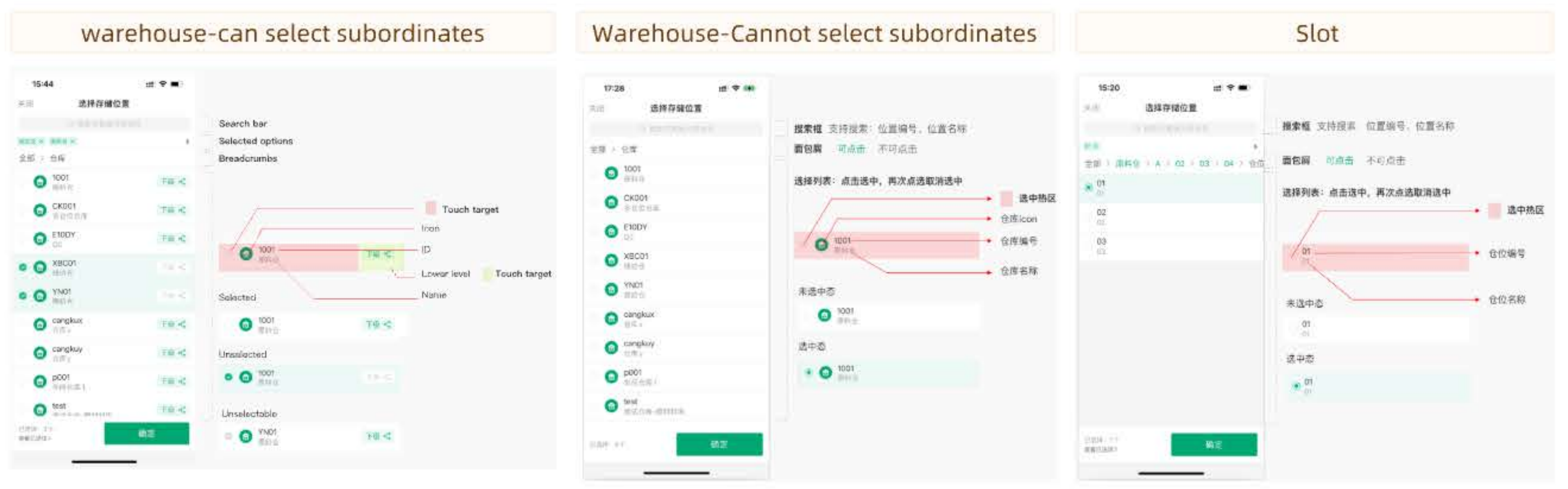


After

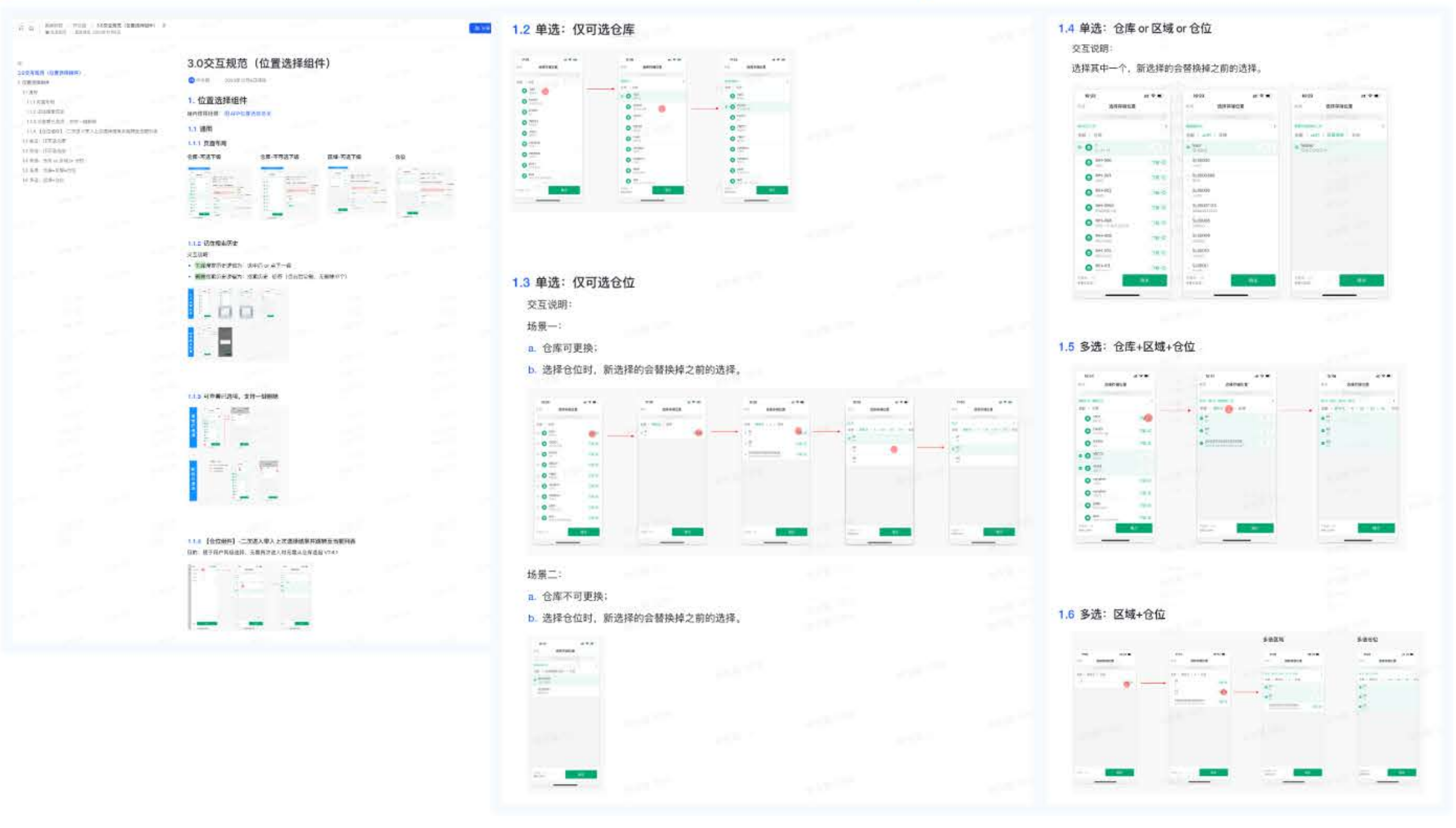
Transform the linearity of the 'lower level' btn into planarity, enhance the click sensation, and expand the touch target. And develop touch target specifications as **design guidelines.**



Develop the page layout



Design interactive drafts and corresponding design guidelines



11 Challenge

Q1: How can I iterate daily requirements and this project in parallel with limited resources (human and time)?

A: Define issues' types and priorities based on the number of participating roles

Priorities	PM	Designer	Front-end
S1-i	🔴	🔴	🟢
S1-i	🔴	🟢	🟢
S1-ii	🟢	🟢	🟢

1. **S1-Phase 1:**
   a. Task operation efficiency、Usability
   b. Task operation efficiency、Usability（The functional and business level solutions need to be improved)

*PMs prioritize individual needs based on their priority

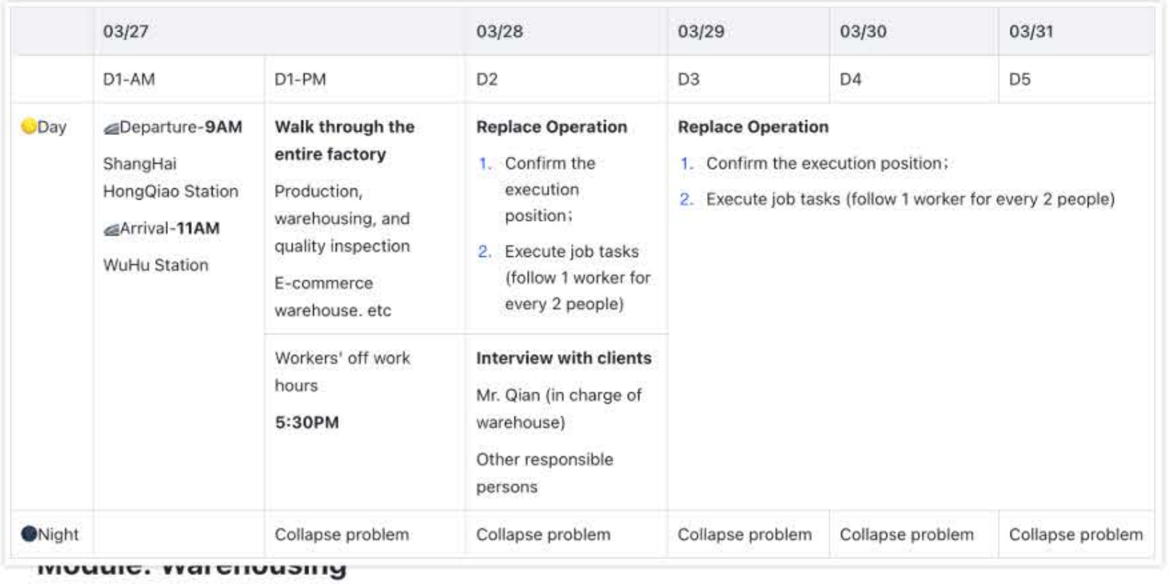
Using iteration as the dimension, in batches (in the most recent iteration, positioned in the review plan)

2. **S2-Phase 2：**
   a. Overall page layout and UI upgrade

Q2: How to effectively discover real and specific problems?

Field research | Replace Operation

A: Assist team members in efficiently anchoring operational issues：
Replace Operation > Interview in the meeting room



	03/27		03/28	03/29	03/30	03/31
	D1-AM	D1-PM	D2	D3	D4	D5
Day	Departure-**9AM** ShangHai HongQiao Station; Arrival-**11AM** WuHu Station	**Walk through the entire factory** Production, warehousing, and quality inspection; E-commerce warehouse. etc	**Replace Operation** 1. Confirm the execution position； 2. Execute job tasks (follow 1 worker for every 2 people)	**Replace Operation** 1. Confirm the execution position； 2. Execute job tasks (follow 1 worker for every 2 people)		
		Workers' off work hours **5:30PM**	**Interview with clients** Mr. Qian (in charge of warehouse); Other responsible persons			
Night		Collapse problem	Collapse problem	Collapse problem	Collapse problem	Collapse problem

Module: Warehousing

Pre-trip Alignment

a. U-Play has multiple factories in Wuhu City, Which one should we visit? Will we be visiting multiple factories?
b. What is the factory's work schedule? What are the morning and afternoon entry times?
c. What is the order of the tour? Do we need to tour the entire factory again the next day?
d. Are there any taboos during the factory visit (places not to visit, questions not to ask)?
e. Provide/coordinate interview personnel (needs the corresponding person in charge/manager for production and warehousing (raw materials, line work, finished goods)...)
f. During the focused factory visit on D2, will everyone go to the same warehouse together, or will we be divided into groups?
g. Replace operations on D3-D5.

12 Review and reflection

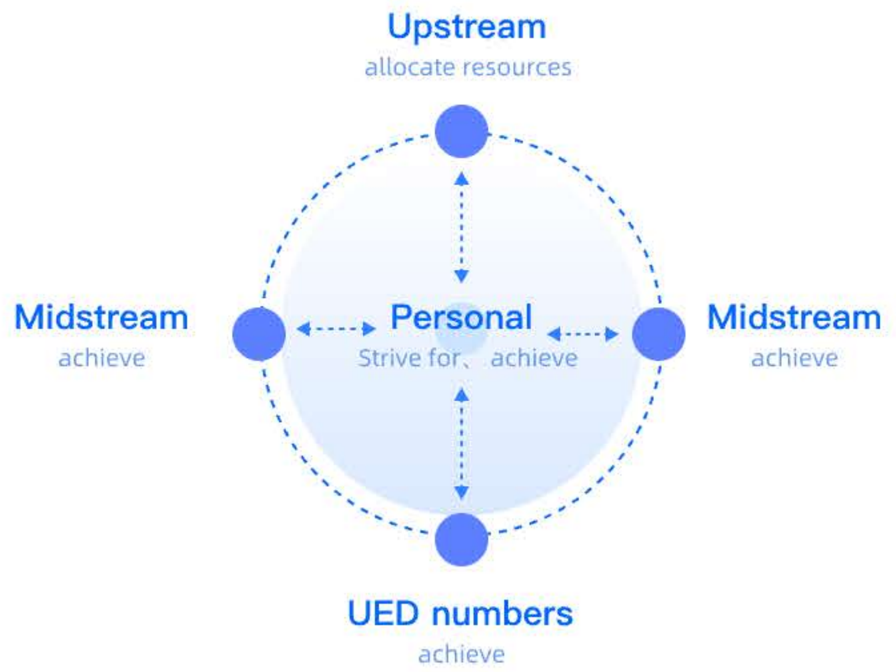
As mentioned at the beginning, the UED Project was established in response to the 930 Campaign.
But why was it given the P0（highest priority)? I tried to review the situation and find an answer.

Seize opportunities

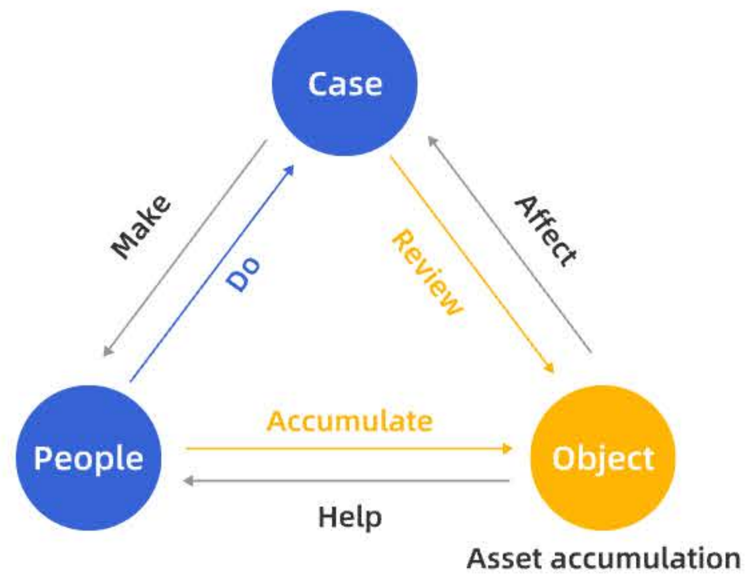
When I first joined Black Lake, I discovered that the entire product and R&D team was technology-oriented and didn't prioritize the importance of experience design. This led to numerous issues during the rapid launch of the product, along with feedback from the customer support team (CS), and during personal use.

Thanks to my then-**teammate Lei(TeamLeader of front-end engineer),** our ideas clicked. After our **initial collaboration on the 2022 HD project**, we developed a mutual trust.
After several rounds of discussions, the GM of the Black Lake Intelligent Manufacturing business line realized that experience issues were a pressing issue that needed to be addressed.

Thinking and collaboration



Professional beliefs as a designer



Asset accumulation

Workers have higher requirements for touch target and color distinguishability.
I have also established: 1. Touch Target Guidelines; 2. System primary color proofreading (as design references for designers and front-end engineers).

1. Touch Target Guidlins | 2. Recalibrated the system's primary color

Using research tools

We don't need to pursue comprehensive methods, instead, we should know how to prescribe the right remedy for the problem. Less is more.
eg: **User Journey Map**

I Shared the UJM's usage experience at the weekly meeting